

Annotation Reference Guide

Spatial CRT Manuscript — Revisions_V1.tex vs. main.tex

BMC Medical Research Methodology — Revision Round 1

This document lists every substantive change made in Revisions_V1.tex relative to the original main.tex. Each entry gives the exact line number(s) in Revisions_V1.tex, the document location, the reviewer comment it addresses, and a description of what was changed.

Summary of All Changed Regions

18 distinct regions changed across 22 revision items. Lines refer to Revisions_V1.tex.

Change ID	Lines	Location in Document	Reviewer Tag	Change Type
R2.M2	283	Section 2.3 — β interpretation paragraph	<i>Reviewer 2, Minor #2</i>	Sentence added
R2.M3	347–349	Figure 2 caption	<i>Reviewer 2, Minor #3</i>	Caption rewritten
R2.M6 / BSS intro	356–372	Section 2.2.1 — BSS description paragraph	<i>Reviewer 2, Minor #6</i>	Paragraph rewritten
R1.1 / R2.2 / R2.3 / R1.4 — para 1	513–515	Introduction (end of Section 1) — first new paragraph	<i>Reviewer 1 Major #1 & #4; Reviewer 2 Major #2 & #3</i>	New paragraph
R1.1 / R2.2 / R2.3 / R1.4 — para 2	516–522	Introduction (end of Section 1) — second new paragraph	<i>Reviewer 1 Major #1 & #4; Reviewer 2 Major #2 & #3</i>	New paragraph
R1.2	538–553	Before Section 2.6 (Spatial Weights Matrix)	<i>Reviewer 1, Major #2</i>	New paragraph
SUPP-1 bridge	579–582	Section 2.6 — spatial weights	<i>Reviewer 2, Major #1 (SUPP-1)</i>	Bridge sentence (equations moved to Appendix A)
SUPP-2 bridge	590–597	Section 2.7 — SAR log-likelihood	<i>Reviewer 2, Major #1 (SUPP-2)</i>	Bridge sentence (equations moved to Appendix B)
R1.3	669–681	Before Section 3.2 (Simulation Design)	<i>Reviewer 1, Major #3</i>	New paragraph
SUPP-3 bridge	676–683	Section 3.2 — simulation algorithm	<i>Reviewer 2, Major #6 (SUPP-3)</i>	Bridge paragraph (8-step algorithm moved to Appendix C)
R2.5	756–757	Section 3.3 — MSE decomposition	<i>Reviewer 2, Minor #5</i>	Sentence added
R1.5 (2×4 grid)	763–775	Section 3.4 — 2×4 results	<i>Reviewer 1, Major #5</i>	Section rewritten (risk-management framing)

R1.5 (3×4 grid) + R2.M4	986–996	Section 3.6 — 3×4 results	<i>Reviewer 1, Major #5; Reviewer 2, Minor #4</i>	Section rewritten (risk-management framing + checkerboard paradox)
R2.4	1189–1201	Section 4.4 — Application (North Carolina)	<i>Reviewer 2, Major #4</i>	Paragraph added to Application section
R2.M4	1212–1218	Section 5.1 — Discussion	<i>Reviewer 2, Minor #4</i>	New paragraph
R1.M2	1220–1222	Section 5.2 — Limitations	<i>Reviewer 1, Minor #2</i>	New paragraph
R1.M3	1224–1228	Section 5.2 — Limitations	<i>Reviewer 1, Minor #3</i>	New paragraph
E1 — Declarations	1266–1292	Declarations (after Conclusion)	<i>Editor Comment 1</i>	Section restructured
SUPP-1–5 — Appendices A–E	1298–1453	Appendices A–E (after Bibliography)	<i>Reviewer 2, Major #1 (SUPP-1–5)</i>	Five entirely new appendix sections

Detailed Change Descriptions

Each entry below gives: line number(s), full location, description of the change, and the first sentence(s) of the changed content.

1. R2.M2 — Sentence added

Lines:	283
Location:	Section 2.3 — β interpretation paragraph
Reviewer:	<i>Reviewer 2, Minor #2</i>
Change:	Added: " β is more precisely interpreted as a total treatment effect rather than a pure direct effect" (with citation) at end of paragraph about β estimation.
Starts with:	<i>...in such scenarios β^* may capture elements of both the direct intervention effect and the indirect spillover...</i>

2. R2.M3 — Caption rewritten

Lines:	347–349
Location:	Figure 2 caption
Reviewer:	<i>Reviewer 2, Minor #3</i>
Change:	Figure 2 caption updated to explicitly label (a) TrtNoSpill and (b) TrtSpill with directional descriptions of spillover arrows.
Starts with:	<i>(a) TrtNoSpill: Spillover arrows flow outward from intervention cells (red) into neighboring control cells (blue) only...</i>

3. R2.M6 / BSS intro — Paragraph rewritten

Lines:	356–372
Location:	Section 2.2.1 — BSS description paragraph
Reviewer:	<i>Reviewer 2, Minor #6</i>
Change:	BSS described as "analogous to geographically stratified randomization in standard CRTs". Added McCann et al. and Crisp et al. citations. Explained novel Rook-edge isolation constraint. Addressed odd-dimension relaxation rule.
Starts with:	<i>Block stratified sampling ensures a balanced number of intervention and control cells are selected, with the added condition...</i>

4. R1.1 / R2.2 / R2.3 / R1.4 — para 1 — New paragraph

Lines:	513–515
Location:	Introduction (end of Section 1) — first new paragraph
Reviewer:	<i>Reviewer 1 Major #1 & #4; Reviewer 2 Major #2 & #3</i>
Change:	Added paragraph situating BSS within spatially balanced experimental design literature. Cites McCann et al. (2018). Articulates novelty: no prior work has compared SRS vs BSS under SAR + spillover with exhaustive enumeration.
Starts with:	<i>The spatial separation principle underlying BSS has antecedents in spatially balanced experimental designs. For example, McCann et al. (2018)...</i>

5. R1.1 / R2.2 / R2.3 / R1.4 — para 2 — New paragraph

Lines:	516–522
Location:	Introduction (end of Section 1) — second new paragraph
Reviewer:	<i>Reviewer 1 Major #1 & #4; Reviewer 2 Major #2 & #3</i>
Change:	Added paragraph distinguishing BSS from covariate-constrained randomization (Crisp et al. 2023). Both restrict allocation space, but on different principles; described as complementary approaches.
Starts with:	<i>A related line of work concerns covariate-constrained randomization, in which the set of eligible allocations is restricted to those...</i>

6. R1.2 — New paragraph

Lines:	538–553
Location:	Before Section 2.6 (Spatial Weights Matrix)
Reviewer:	<i>Reviewer 1, Major #2</i>
Change:	Added paragraph clarifying conceptual distinction between ρ (nuisance background autocorrelation) and ψ (causal design-level spillover). Explains why failing to separate them can produce confounded estimates.
Starts with:	<i>Before describing the spatial weights matrix, it is important to clarify the conceptual distinction between two spatial quantities...</i>

7. SUPP-1 bridge — Bridge sentence (equations moved to Appendix A)

Lines:	579–582
Location:	Section 2.6 — spatial weights
Reviewer:	<i>Reviewer 2, Major #1 (SUPP-1)</i>
Change:	Equations for W and w_{ij} moved to Appendix A. Main body retains bridge sentence summarizing W construction with pointer to Appendix A.
Starts with:	<i>The spatial weights matrix W encodes the strength of spatial connections between observations; see Appendix A for full formulation.</i>

8. SUPP-2 bridge — Bridge sentence (equations moved to Appendix B)

Lines:	590–597
Location:	Section 2.7 — SAR log-likelihood
Reviewer:	<i>Reviewer 2, Major #1 (SUPP-2)</i>
Change:	Full SAR log-likelihood formula moved to Appendix B. Subject-level estimating equation retained in main body; full likelihood in Appendix B.
Starts with:	<i>Maximum likelihood estimation is utilized to gather parameter estimates from the spatial lag model...</i>

9. R1.3 — New paragraph

Lines:	669–681
Location:	Before Section 3.2 (Simulation Design)
Reviewer:	<i>Reviewer 1, Major #3</i>
Change:	Added paragraph clarifying inferential scope: BSS is a design-stage tool; exhaustive enumeration characterizes design space; model-based inference via SAR MLE is appropriate; simulation findings describe how each design tends to perform across feasible allocations.
Starts with:	<i>Before describing the simulation procedure, it is important to clarify the inferential scope of this study...</i>

10. SUPP-3 bridge — Bridge paragraph (8-step algorithm moved to Appendix C)

Lines:	676–683
Location:	Section 3.2 — simulation algorithm
Reviewer:	<i>Reviewer 2, Major #6 (SUPP-3)</i>
Change:	8-step simulation algorithm enumeration moved to Appendix C. Main body retains compact summary paragraph with pointer to Appendix C.
Starts with:	<i>For each spatial grid geometry and simulation scenario, the complete set of possible treatment combinations was enumerated exhaustively...</i>

11. R2.5 — Sentence added

Lines:	756–757
Location:	Section 3.3 — MSE decomposition
Reviewer:	<i>Reviewer 2, Minor #5</i>
Change:	Added: "Since $MSE = Bias^2 + Variance$, differences in MSE between SRS and BSS can be attributed to either systematic bias or sampling variability across allocations."
Starts with:	<i>Since $MSE = Bias^2 + Variance$, differences in MSE between SRS and BSS can be attributed to either systematic bias or sampling variability...</i>

12. R1.5 (2×4 grid) — Section rewritten (risk-management framing)

Lines:	763–775
Location:	Section 3.4 — 2×4 results
Reviewer:	<i>Reviewer 1, Major #5</i>
Change:	Sections 3.4 rewritten: introduced risk-management framing. BSS "bounds the worst-case outcome"; SRS is a "lottery" offering lower average MSE but with high variance across 70 allocations. SD comparison emphasized throughout.
Starts with:	<i>...reveal a risk-management trade-off between SRS and BSS rather than a uniform advantage for either method...</i>

13. R1.5 (3×4 grid) + R2.M4 — Section rewritten (risk-management framing + checkerboard paradox)

Lines:	986–996
Location:	Section 3.6 — 3×4 results
Reviewer:	<i>Reviewer 1, Major #5; Reviewer 2, Minor #4</i>
Change:	Section 3.6 rewritten. Introduced "checkerboard paradox": under TrtSpill + $\rho=0.01$, BSS MSE (≈ 0.223) substantially exceeds SRS MSE (0.112) because checkerboard surrounds every intervention cluster exclusively with controls, amplifying spillover bias.
Starts with:	<i>...reveal both the typical strengths of BSS and a notable exception that illuminates the limits of the checkerboard design...</i>

14. R2.4 — Paragraph added to Application section

Lines:	1189–1201
Location:	Section 4.4 — Application (North Carolina)
Reviewer:	<i>Reviewer 2, Major #4</i>
Change:	Added paragraph noting that real-world BSS is "best-effort": irregular district shapes prevent perfect spatial separation, and investigators should document the degree to which their chosen allocation approximates the strict criterion.
Starts with:	<i>...the allocation depicted therefore represents a best-effort block stratification: the assignment minimizes shared borders...</i>

15. R2.M4 — New paragraph

Lines:	1212–1218
Location:	Section 5.1 — Discussion
Reviewer:	<i>Reviewer 2, Minor #4</i>
Change:	Added paragraph in Discussion explaining the checkerboard paradox: BSS's spatial isolation feature becomes a liability under TrtSpill + $\rho > 0$. Recommendation: consider prior knowledge of spillover mechanism when selecting SRS vs BSS.
Starts with:	<i>An important nuance observed in the 3×4 simulation results warrants attention in the context of practical design recommendations...</i>

16. R1.M2 — New paragraph

Lines:	1220–1222
Location:	Section 5.2 — Limitations
Reviewer:	<i>Reviewer 1, Minor #2</i>
Change:	Added limitation paragraph: spillover modeled as binary (full effect or none, contiguity-based only). Does not capture distance decay, additive effects from multiple treated neighbors, or non-linear patterns.
Starts with:	<i>An additional notable limitation is the rudimentary spillover mechanism (binary effect) applied...</i>

17. R1.M3 — New paragraph

Lines:	1224–1228
Location:	Section 5.2 — Limitations
Reviewer:	<i>Reviewer 1, Minor #3</i>
Change:	Added limitation paragraph: findings characterize design-space performance via exhaustive enumeration, not a sampling distribution from a single randomization. Simulation is not designed to support permutation inference under BSS.
Starts with:	<i>Because this study evaluates estimator properties through exhaustive enumeration of all possible treatment allocations...</i>

18. E1 — Declarations — Section restructured

Lines:	1266–1292
Location:	Declarations (after Conclusion)
Reviewer:	<i>Editor Comment 1</i>
Change:	Entire Declarations section rebuilt with 7 BMC-standard subsections: Ethics Approval, Consent for Publication, Data Availability, Competing Interests, Funding, Authors' Contributions, Acknowledgements.
Starts with:	<i>\section*{Declarations} → 7 \subsection* entries per BMC Medical Research Methodology style</i>

19. SUPP-1–5 — Appendices A–E — Five entirely new appendix sections

Lines:	1298–1453
Location:	Appendices A–E (after Bibliography)
Reviewer:	<i>Reviewer 2, Major #1 (SUPP-1–5)</i>
Change:	Appendix A: Spatial Weights Matrix (from §2.6). Appendix B: SAR Full Likelihood (from §2.7). Appendix C: Simulation Algorithm (from §3.2). Appendix D: Moran's I and Geary's C (from §2.6). Appendix E: Distance-Based Neighbor Construction (from §2.2.2).
Starts with:	<i>Lines 1298–1453 are entirely new in Revisions_V1.tex (did not exist in main.tex)</i>